

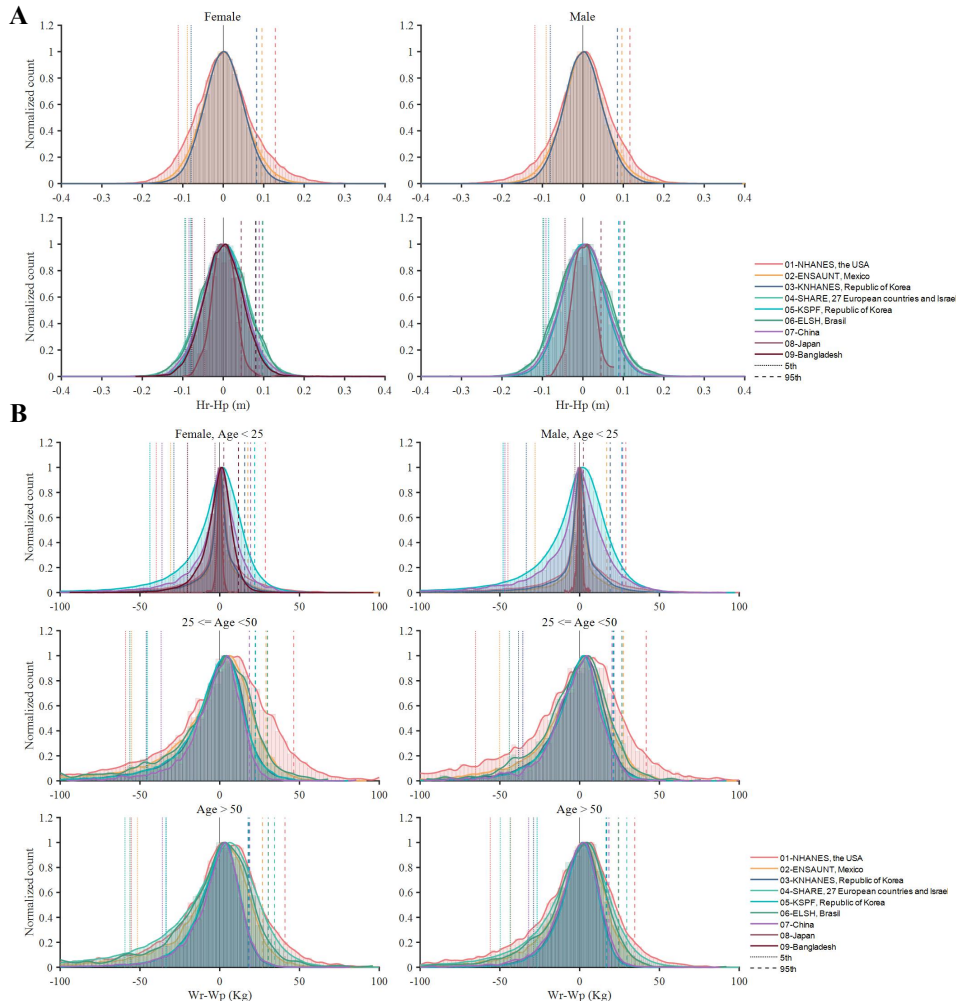


Fig. 5. Distribution curves for weight and height residuals. (A) Distribution curves for height residual. (B) Distribution curves for weight residual. To elucidate age-dependent patterns, the weight residuals were categorized into three age groups: Age < 25 years (top two panels), $25 \leq \text{Age} < 50$ years (middle two panels), and Age ≥ 50 years (bottom two panels). To prevent predicted weight approaching infinity as height nears the theoretical upper limit ($1/b$), an exclusion criterion was applied to omit data points with weight residuals falling outside the biologically plausible range of ± 100 kg. The 5th and 95th percentile positions are marked to illustrate symmetry characteristics. The weight and height residuals (Hr-Hp, Wr-Wp) are defined as the differences between the observed values and the constraint-model-predicted values.